

## WIELUN, Poland

WMO: 124550

Lat: 51.2102N

Lon: 18.5566E

Elev: 201

StdP: 98.94

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.0      | -10.9 | -16.9                         | 0.9 | -13.2 | -14.1 | 1.1 | -9.9 | 9.9                   | 4.4  | 8.9 | 3.8  | 2.4                   | 90   | 0.515 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.9                    | 30.1            | 19.7 | 28.2 | 18.9 | 26.5 | 18.2 | 20.8                | 27.5 | 20.0 | 26.1 | 19.2 | 24.8 | 2.9                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.7                            | 13.9 | 23.2 | 17.9 | 13.2 | 22.5 | 17.1 | 12.5 | 21.5 | 61.0          | 27.7 | 58.1 | 26.2 | 55.3 | 24.9 | 26.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 7.9           | 6.8  | 6.0 |                            | -17.2 | 33.6               | 3.8 | 1.6                                                | -19.9 | 34.7       | -22.2 | 35.7       | -24.3 | 36.5       | -27.1 | 37.7 |     |
| (5)               |      |     | DB                         | -17.5 | 22.8               | 3.7 | 1.2                                                | -20.2 | 23.6       | -22.4 | 24.3       | -24.5 | 24.9       | -27.2 | 25.8 |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.1    | -1.4  | 0.0   | 3.6   | 9.3   | 13.9  | 17.1  | 19.2  | 18.9  | 14.3  | 9.4   | 4.6   | 0.3   |
|                                                                        | DBStd       | 8.36   | 5.20  | 4.97  | 4.40  | 4.36  | 3.91  | 3.54  | 3.31  | 3.14  | 3.52  | 4.27  | 4.19  | 4.70  |
|                                                                        | HDD10.0     | 1443   | 352   | 280   | 201   | 64    | 9     | 0     | 0     | 0     | 4     | 64    | 167   | 301   |
|                                                                        | HDD18.3     | 3499   | 611   | 513   | 456   | 272   | 145   | 65    | 29    | 32    | 127   | 277   | 413   | 559   |
|                                                                        | CDD10.0     | 1131   | 0     | 0     | 3     | 42    | 131   | 212   | 284   | 274   | 134   | 46    | 4     | 0     |
|                                                                        | CDD18.3     | 146    | 0     | 0     | 0     | 1     | 8     | 27    | 56    | 48    | 7     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 1341   | 0     | 0     | 0     | 16    | 87    | 258   | 516   | 404   | 59    | 1     | 0     | 0     |
|                                                                        | CDH26.7     | 352    | 0     | 0     | 0     | 1     | 11    | 67    | 158   | 104   | 11    | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 2.9    | 3.5   | 3.4   | 3.4   | 3.1   | 2.7   | 2.5   | 2.5   | 2.4   | 2.6   | 2.9   | 3.2   | 3.5   |
| (15) Precipitation                                                     | PrecAvg     | 605    | 36    | 32    | 39    | 37    | 71    | 73    | 92    | 54    | 52    | 39    | 42    | 38    |
|                                                                        | PrecMax     | 842    | 80    | 59    | 96    | 82    | 213   | 150   | 315   | 124   | 135   | 90    | 87    | 119   |
|                                                                        | PrecMin     | 390    | 7     | 5     | 15    | 1     | 27    | 23    | 34    | 15    | 17    | 4     | 0     | 10    |
|                                                                        | PrecStd     | 110    | 15    | 15    | 20    | 18    | 39    | 34    | 61    | 25    | 27    | 22    | 18    | 21    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 10.4  | 12.1  | 17.8  | 25.5  | 28.0  | 31.6  | 33.1  | 32.1  | 28.2  | 22.5  | 15.6  | 11.3  |
|                                                                        |             | MCWB   | 7.8   | 8.4   | 10.6  | 15.5  | 17.7  | 20.9  | 20.3  | 20.2  | 17.6  | 15.8  | 11.9  | 9.0   |
|                                                                        | 2%          | DB     | 8.1   | 9.9   | 14.7  | 21.9  | 25.8  | 28.5  | 30.5  | 29.3  | 24.9  | 20.0  | 13.6  | 9.2   |
|                                                                        |             | MCWB   | 6.3   | 7.3   | 9.2   | 13.4  | 17.0  | 19.7  | 19.7  | 19.5  | 16.9  | 14.9  | 10.8  | 7.6   |
|                                                                        | 5%          | DB     | 6.5   | 8.1   | 12.1  | 19.1  | 23.7  | 26.3  | 28.3  | 27.4  | 22.8  | 17.6  | 11.9  | 7.5   |
|                                                                        |             | MCWB   | 5.0   | 6.1   | 8.1   | 11.8  | 16.1  | 18.4  | 19.1  | 18.8  | 16.0  | 13.7  | 9.7   | 6.2   |
|                                                                        | 10%         | DB     | 5.1   | 6.5   | 10.0  | 16.9  | 21.5  | 24.2  | 26.4  | 25.5  | 20.7  | 15.5  | 10.3  | 6.0   |
|                                                                        |             | MCWB   | 3.9   | 4.7   | 7.0   | 10.8  | 14.9  | 17.2  | 18.3  | 18.0  | 15.2  | 12.6  | 8.6   | 4.8   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 8.2   | 8.9   | 11.4  | 16.1  | 19.3  | 22.0  | 21.9  | 21.6  | 18.7  | 16.9  | 12.4  | 9.4   |
|                                                                        |             | MCDB   | 9.8   | 10.5  | 16.1  | 23.4  | 25.7  | 29.1  | 29.3  | 28.2  | 24.7  | 20.7  | 14.7  | 11.0  |
|                                                                        | 2%          | WB     | 6.5   | 7.6   | 9.8   | 14.2  | 17.9  | 20.5  | 20.9  | 20.4  | 17.7  | 15.3  | 11.2  | 7.7   |
|                                                                        |             | MCDB   | 7.9   | 9.7   | 13.6  | 20.6  | 23.6  | 27.1  | 27.5  | 27.1  | 23.1  | 18.7  | 13.4  | 9.0   |
|                                                                        | 5%          | WB     | 5.1   | 6.2   | 8.5   | 12.7  | 16.8  | 19.3  | 20.1  | 19.6  | 16.8  | 14.1  | 9.9   | 6.3   |
|                                                                        |             | MCDB   | 6.4   | 8.1   | 11.6  | 17.9  | 22.3  | 24.8  | 26.0  | 25.7  | 21.6  | 17.1  | 11.6  | 7.4   |
|                                                                        | 10%         | WB     | 3.9   | 4.7   | 7.2   | 11.4  | 15.7  | 18.1  | 19.2  | 18.8  | 15.7  | 12.8  | 8.7   | 4.9   |
|                                                                        |             | MCDB   | 5.0   | 6.3   | 9.9   | 15.8  | 20.6  | 23.1  | 24.7  | 24.0  | 19.6  | 15.2  | 10.1  | 5.9   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 4.5   | 5.4   | 7.1   | 9.4   | 9.9   | 9.8   | 9.9   | 10.1  | 8.7   | 7.2   | 5.1   | 4.2   |
|                                                                        |             | MCDBR  | 5.1   | 7.3   | 10.8  | 13.5  | 13.3  | 13.4  | 13.5  | 13.2  | 12.2  | 10.1  | 7.5   | 5.2   |
|                                                                        |             | MCWBR  | 4.3   | 5.3   | 6.6   | 6.7   | 6.4   | 6.2   | 5.0   | 5.3   | 5.5   | 5.8   | 5.2   | 4.2   |
|                                                                        | 5% WB       | MCDBR  | 5.1   | 7.0   | 9.7   | 12.0  | 11.9  | 12.1  | 11.5  | 11.6  | 10.5  | 9.4   | 7.1   | 5.1   |
|                                                                        |             | MCWBR  | 4.3   | 5.4   | 6.4   | 6.2   | 6.1   | 6.2   | 4.9   | 5.1   | 5.3   | 5.7   | 5.2   | 4.2   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.316 | 0.343 | 0.381 | 0.431 | 0.424 | 0.420 | 0.446 | 0.434 | 0.399 | 0.371 | 0.347 | 0.312 |
|                                                                        | taud        |        | 2.281 | 2.271 | 2.216 | 2.135 | 2.207 | 2.251 | 2.191 | 2.228 | 2.295 | 2.337 | 2.338 | 2.324 |
|                                                                        | Ebn at Noon |        | 685   | 769   | 805   | 805   | 831   | 838   | 806   | 795   | 780   | 728   | 643   | 630   |
|                                                                        | Edh at Noon |        | 68    | 90    | 114   | 139   | 135   | 131   | 137   | 125   | 105   | 84    | 65    | 56    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.76  | 1.47  | 2.66  | 4.12  | 5.11  | 5.37  | 5.05  | 4.53  | 3.24  | 1.86  | 0.87  | 0.58  |
|                                                                        | RadStd      |        | 0.14  | 0.26  | 0.34  | 0.54  | 0.56  | 0.53  | 0.61  | 0.38  | 0.45  | 0.28  | 0.12  | 0.08  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | +0.45   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (29 neighbors) |       | +0.50   | N/A    | N/A     | +0.74 | N/A   | N/A         | -163    | +74     | +27     |

Nomenclature: See separate page